H.T.No: | | | | | | | | | | |

Course Code No: 23CS3T01

VISHNU INSTITUTE OF TECHNOLOGY (AUTONOMOUS)

II B. Tech I Semester (R23) Regular Examinations, November– 2024

DATABASE MANAGEMENT SYSTEMS

(CSE, IT, AI & DS, AI & ML and CSBS)

Time:3 Hours **Max. Marks: 70M**

Note:1. Question Paper consists of two parts (**Part-A** and **Part-B**)
2. Answer all the questions in **Part-A**
3. Answer either a or b questions from Question number 2 to 6 in **Part-B**

PART –A

1	I	Discuss subclasses and super class in ER Model.	L2	CO1	[2M
	II	What is an entity? Explain with example.	L2	CO1	[2M
	III	Write about the importance of null values.	L2	CO2	[2M
	IV	What is domain constraint?	L1	CO2	[2M
	V	What is a nested query?	L1	CO3	[2M
	VI	Distinguish between left and right outer join.	L3	CO3	[2M
	VII	What are the problems caused by redundancy in database design?	L1	CO4	[2M
	VIII	What is surrogate key?	L1	CO4	[2M
	IX	What is deadlock?	L1	CO5	[2M
	X	What is a serializable schedule?	L1	CO5	[2M

PART –B

UNIT – 1

2(a) In the following ER Diagram identify entities, relationship, mapping cardinality, and participation constraints. Explain each of them. L1 CO1 [10

(OR)

2(b) *I.* Explain different types of data independence. L2 CO1 [4M

II. Draw an ER diagram for the following information of Vishnu Institute of Technology. All students register for various courses. Student information such as roll number, name, address and course information such as course L3 CO1 [6M

id, name and credits for the course are saved. Indicate attributes, entity, primary keys, relationship, partial and total participation in ER diagram.

UNIT – II

3(a)		Write the properties of Tuple Relation Calculus. Explain with an example.	L2	CO2 [10M]
		(OR)		
3(b)	I	*Create a query considering two relations, Students (Pin, name, course) and Register (sid, dept, course), with 'sid' as the foreign key in Students referencing the pin as primary key in Register. What are the violations can cause of the referential integrity constraint? Why?*	L3	CO2 [6M]
	II	*Explain any 2 DML commands with suitable example.*	L2	CO2 [4M]

UNIT – III

4(a)		*What is the view? Describe the difference between updatable and non-updatable views.*	L2	CO3 [10M]
		(OR)		
4(b)		Explain the concept of aggregation functions in SQL. Discuss how min aggregate function is used with the nested query with example.	L2	CO3 [10M]

UNIT – IV

5(a)		Consider schema R = (A, B, C, G, H, I) and the set F of functional dependencies $\{A \rightarrow B, A \rightarrow C, CG \rightarrow H, CG \rightarrow I, B \rightarrow H\}$. Compute the candidate keys of the schema. The given relation is in which normal form and why?	L3	CO4 [10M]
		(OR)		
5(b)	I	What is decomposition and discuss its properties.	L1	CO4 [7M]
	II	Explain the concept of multivalued dependency.	L2	CO4 [3M]

UNIT – V

6 (a)		Explain hash-based indexing and its key characteristics. How does it differ from other indexing methods?	L2	CO5 [10M]
		(OR)		
6(b)	I	Differentiate between conflict serializable and view serializable schedules.	L3	CO5 [6M]
	II	Explain ACID properties of Transactions.	L2	CO5 [4M]

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